

ON FEBRUARY 10, 2023, when the Geological Survey of India announced the 'discovery' of lithium ore in Jammu & Kashmir, the whole country celebrated. After all, lithium is the fuel behind the fuel (batteries) of electric vehicles, smartphones and any other type of device that uses a rechargeable battery today. And India is importing most of its requirements from, guess who? China, of course. So, now, we have lithium in our own country! We can make our own batteries, save on our import bill, and do a lot of other things.

Well, don't pop the champagne just yet. For one, the lithium found in J&K was actually first discovered in 1999. And even though it is used in industries like speciality chemicals, glass, etc., India has continued to import the soft metal because mining it domestically is very capital intensive and not entirely profitable. Well, those caveats still exist, even though the exponential rise in demand would surely soften them. But the bigger challenge now is that of time and money—time that would be needed to first understand how much of the reserves can be commercially mined, and then the time it'll take to get to the mining stage; and the bagfuls of money that would need to be pumped in to build a completely new ecosystem of industries to leverage the metal.

Often called 'white gold', lithium has emerged as a priority metal for India given its ambitious target of achieving 30 per cent sales of EVs, along with increasing non-fossil fuel energy capacity to 500 gigawatt by 2030. "India's annual lithium-ion battery market is expected to grow to 116 GWh (gigawatt hour) by FY30, from 2.6 GWh in FY21, with EVs accounting for



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TURNING STONE INTO GOLD

General steps involved in getting lithium from its ore

EXPLORATION

A potential lithium deposit is identified, and geological surveys are conducted to assess the size, location and quality of the deposit

DEVELOPMENT A mining firm is assigned to develop the site; this may involve building access roads, constructing buildings and infrastructure, and installing equipment

EXTRACTION This involves extracting the ores from the mine using heavy

machinery to dig underground or open-pit mines

CONCENTRATION

The extracted ore is processed to remove impurities and concentrate the lithium using physical or chemical methods, such as crushing, grinding, froth flotation, or dense media separation

CONVERSION

The concentrated lithium is converted into a more usable form, such as lithium carbonate or lithium

hydroxide, involving a series of chemical reactions with additional processing and refining

PURIFICATION

The final step involves purifying the lithium product to remove any remaining impurities

SUPPLY

Once the lithium has been purified, it is sold to manufacturers who use it to make batteries, ceramics, glass and other products

SOURCE: BT RESEARCH

GRAPHICS BY RAHUL SHARMA

90 per cent of the market—according to a report by JMK Research & Analytics and Institute for Energy Economics and Financial Analysis," says Prahalathan Iyer, Chief General Manager for Research & Analysis at India Exim Bank.

"The discovery of lithium (inferred) reserves in J&K, is strategically important as it improves India's energy security by securing the critical mineral's supplies and also builds self-sufficiency," explains Anish Mandal, Partner at Deloitte India, adding that apart from reducing India's import bill and trade deficit, the reserves could also help the country achieve

its net-zero target by accelerating green transportation and green energy adoption. For context, around one tonne of lithium can be used to produce batteries for around 90-120 4-wheeler EVs.

They key operative word here, therefore, is 'inferred'—the discovery is under the 'inferred category' signifying low confidence in the reserves and requiring more detailed studies. So, no, we can't use the J&K lithium just yet.

JUMPING THE GUN

If the reserves turn out to be as massive as claimed, India may find itself joining a list of select nations